

Advanced Coordinate Geometry



Distance, midpoint, and slope combined

- 1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB .
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Equations of circles from given conditions

- 2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.
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Distance from a point to a line

- 3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
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Locus problems

- 4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.
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Rotations and general transformations

- 5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \rightarrow (-y, x)$.
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Dilations and scale factor

- 6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.
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Finding the equation of a parabola from three points

- 7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a , b , and c .
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A multi-part triangle-in-the-plane problem

- 8 This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.
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Finding the equation of a line given a point and a perpendicular condition

- 9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.
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Translating and reflecting equations of curves

- 10 The graph of $y = x^2$ is reflected over the x -axis, then shifted 3 units up. Write the equation of the resulting graph.
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Finding vertices of a polygon given transformations

- 11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.
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Finding the intersection of a line and a parabola

- 12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.
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A multi-part quadrilateral classification problem

- 13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY , and of WZ and XY . (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.
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Finding the equation of a tangent line to a circle

- 14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.
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Finding the vertex and axis of symmetry of a parabola from three points

- 15** A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.